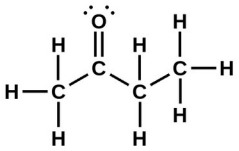
Structure, Bonding, Resonance & Naming: Guide

This guide goes with the Structure & Bonding practice packet. Keep it next to you while you work. Every structure here shows its lone pairs so you can count electrons.

Reading structures

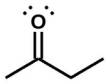
Molecules can be represented many ways, but for now we will be focusing on one main representation, the bond-line structure. Here is butanone three ways:



Full structure
every atom and bond

$CH_3COCH_2CH_3$

Condensed
each C with its H's



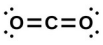
Bond-line
C and H on C left out

1. Every corner and every line end is a carbon.
2. Each carbon has four bonds. Any bonds you don't see are to H.
3. Heteroatoms (O, N, Cl, Br…) are always written out, with the H atoms on them.
4. A neutral O has two lone pairs, a neutral N has one, and a halogen has three. Many books leave them out; I draw them.

Molecular formula: count C, then H, then the other atoms in alphabetical order. Butanone is C_4H_8O.

Hybridization, shape and σ/π bonds

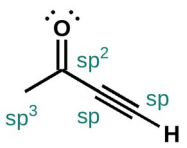
Count the electron groups on the atom. Each bonded atom is one group, even through a double or triple bond, and each lone pair is one group.

Electron groups	Hybridization	Shape of the groups	Bond angle	Example
4	sp^3	tetrahedral	109.5°	
3	sp^2	trigonal planar	120°	
2	sp	linear	180°	

Shortcut for carbon: only single bonds → sp^3; one double bond → sp^2; a triple bond or two double bonds → sp.

The exception: a lone pair next to a π bond is usually part of the π system, so that atom is sp^2 even though you count four groups. The N of an amide is the classic case: it is flat (see DMF under Resonance).

Counting bonds: every bond has one σ. A double bond adds one π, a triple bond adds two. Don't forget the C–H bonds you can't see.



Example: *but-3-yn-2-one has 8 σ and 3 π bonds.*

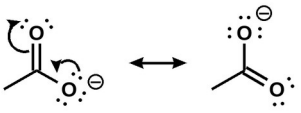
Check it in 3D: the three molecules in the table are in the VSEPR Builder at chemustad.org/vsepr.html. Part 1 of the practice packet walks you through it.

Resonance

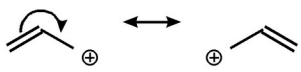
Sometimes one Lewis structure can't describe a molecule. Resonance structures put the same electrons in different places on the same skeleton. The real molecule is a blend (a hybrid) of all of them; it does not flip back and forth.

1. Atoms never move. Only π electrons and lone pairs move.
2. Curved arrows start at a lone pair or a π bond and point to where the electrons go.
3. The total charge stays the same in every structure.
4. C, N and O never have more than 8 electrons.

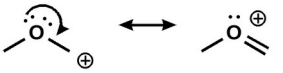
Four patterns cover almost everything you will see:



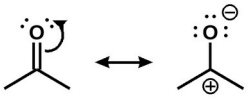
1. lone pair next to a π bond



2. + charge next to a π bond

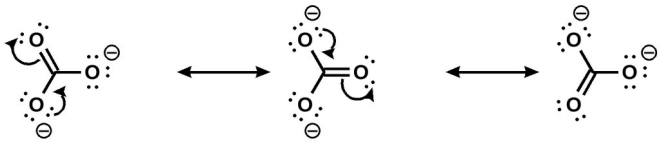


3. lone pair next to a + charge



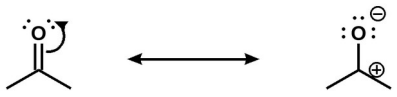
4. π bond to a more electronegative atom

Carbonate. CO_2 dissolves in basic water as bicarbonate and carbonate. The three structures of carbonate are equal, so every C–O bond is about 1⅓ bonds and every O carries about −⅔ charge. This chemistry was the center of my Ph.D. research on capturing CO_2.



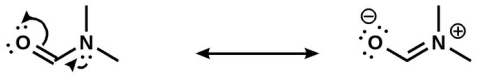
Which structure counts most? Rank unequal structures with these rules, in order:

1. The most atoms with a full octet.
2. The fewest charges.
3. Negative charge on the more electronegative atom (O before N before C); positive charge on the less electronegative one.



major: no charges minor: C has only 6 electrons

Acetone: the neutral structure wins (rule 2). The minor one is why the C=O carbon is partly positive.

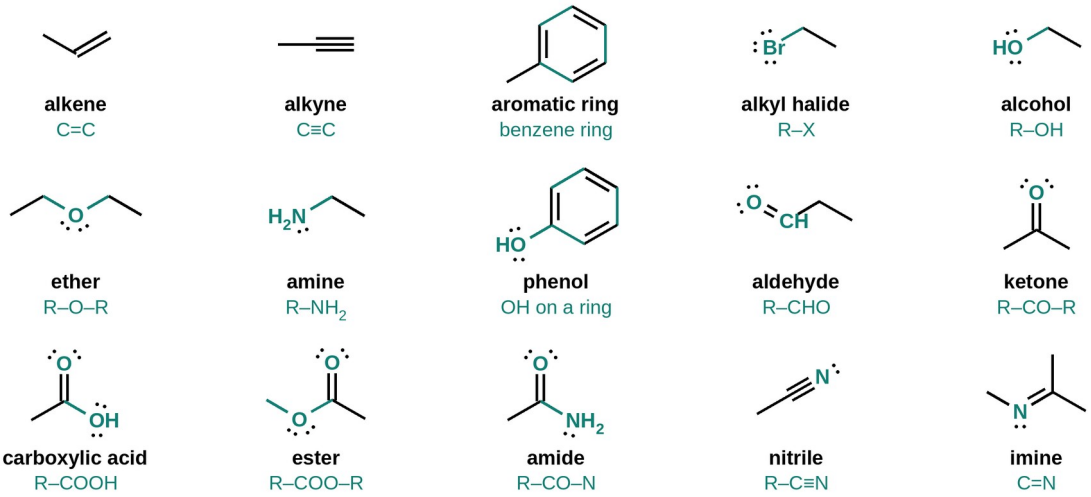


major minor

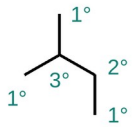
DMF: the minor structure has more charges, but every atom keeps an octet (rule 1), so it matters a lot. That is why the amide C–N bond doesn't rotate freely and the N is flat and sp².

Functional groups and types of carbon

The functional group is the part of the molecule that reacts. In each example the group is drawn in teal.



A carbon is primary (1°), secondary (2°), tertiary (3°) or quaternary (4°) depending on how many other carbons it is bonded to. Alcohols and alkyl halides take the label of the carbon that holds the OH or the halogen.



Degrees of unsaturation and isomers

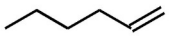
Degrees of unsaturation = (2C + 2 + N − H − X) ÷ 2

Each ring or π bond removes two H. X is the number of halogens; O doesn't change the count. Two degrees could be two rings, two double bonds, one of each, or one triple bond.

Example: *C_6H_{12} has (12 + 2 − 12) ÷ 2 = 1 degree: one ring or one C=C.*



cyclohexane: 1 ring



hex-1-ene: 1 π bond

Constitutional isomers have the same formula but different connections. A chain drawn bent or flipped is still the same molecule, so check for repeats by naming each one.

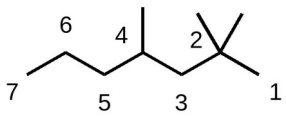
Naming alkanes, cycloalkanes, alkyl halides and alcohols

1. Find the longest carbon chain (the parent): meth-, eth-, prop-, but-, pent-, hex-, hept-, oct-, non-, dec- + "-ane". If two tie, pick the one with more substituents.

2. Number from the end that gives the first substituent the lowest number. If that ties, go to the next one.

3. Name the substituents: methyl, ethyl, propyl, isopropyl, tert-butyl; fluoro, chloro, bromo, iodo.

4. List them in alphabetical order. Use di-, tri-, tetra- for repeats, but ignore those prefixes when you

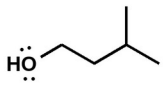


Example: *from the right: 2, 2, 4. From the left: 4, 6, 6. 2,2,4-Trimethylheptane.*

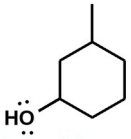
alphabetize.

Rings: the ring is the parent unless a chain on it has more carbons. With one substituent no number is needed; with two, give 1 to the one first in the alphabet unless another numbering gives lower numbers overall.

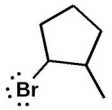
Alcohols: the chain must contain the carbon with the OH, and the OH gets the lowest number. Change -ane to -an-#-ol. The OH outranks alkyl groups and halogens when you number.



3-methylbutan-1-ol



3-methylcyclohexan-1-ol



1-bromo-2-methylcyclopentane

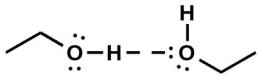
Intermolecular forces

Boiling point and solubility depend on how strongly molecules hold on to each other. From weakest to strongest:

1. London dispersion: in every molecule. Bigger, less branched molecules have more.

2. Dipole–dipole: between polar molecules, such as aldehydes, ketones and alkyl halides.

3. Hydrogen bonds: an H on O, N or F attracted to a lone pair on another O, N or F. Alcohols, carboxylic acids, and 1° and 2° amines. (3° amines have no N–H, so they can only accept one.)



hydrogen bond



pentane, bp 36 °C



2,2-dimethylpropane, bp 10 °C

For molecules of similar size, more hydrogen bonding means a higher boiling point, and more branching means a lower one. “Like dissolves like”: a molecule that can hydrogen-bond with water dissolves better, until its nonpolar carbon chain gets too long.